

THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

A3 BOOKLET

Haidian District, Beijing • both sides of the Jing-Zhang Rail Heritage Park • three key areas: Zhongzhiyuan • Beijing AI Origin Community • Dazhongsi AI Industry Cluster

A civic AI leveling network that uses closure error as its trust metric

Leveling is not about measuring accurately; it is about measuring back.
Depart, carry the height station by station, return — that gap is f.
Within tolerance, every station on the line is accepted;
over tolerance the whole run is void; no station may be patched alone.

That hundred-year-old rule answers this belt's hardest question:
what a city's trust in its AI rests on: not performance, but return.



Ground within a ten-minute walk of a benchmark

Dashed: 232 buildings today. Filled: the ground that only comes within ten minutes once the eleven stitching links and three spurs are built — 261 in all, a gain of 29.
Catchment 203.3 ha to 220.6 ha; 800 m of network distance, not a radius.

READINGS, ALL RECOMPUTED FROM THE SHIPPED FILES

11,412,825 m²

Overall design area

13.8 km/km²

Network density, measured

9,443 m

Spine length

17

Oversized blocks, not parks

8

Tiered benchmarks

19.7%

Green corridor, share of design area

What this proposal does not give here matters as much as what it does

Plot ratio, height, coverage, setbacks, road redlines — inside the statutory approval process; none is drafted here before the official conditions are published.

The tolerance values per scenario — set by the tolerance assembly at BM-1; this proposal gives the grading basis and drafts no value.

Any engineering feasibility conclusion — stitching points register a need for connection; transport and structural review is outside this proposal.

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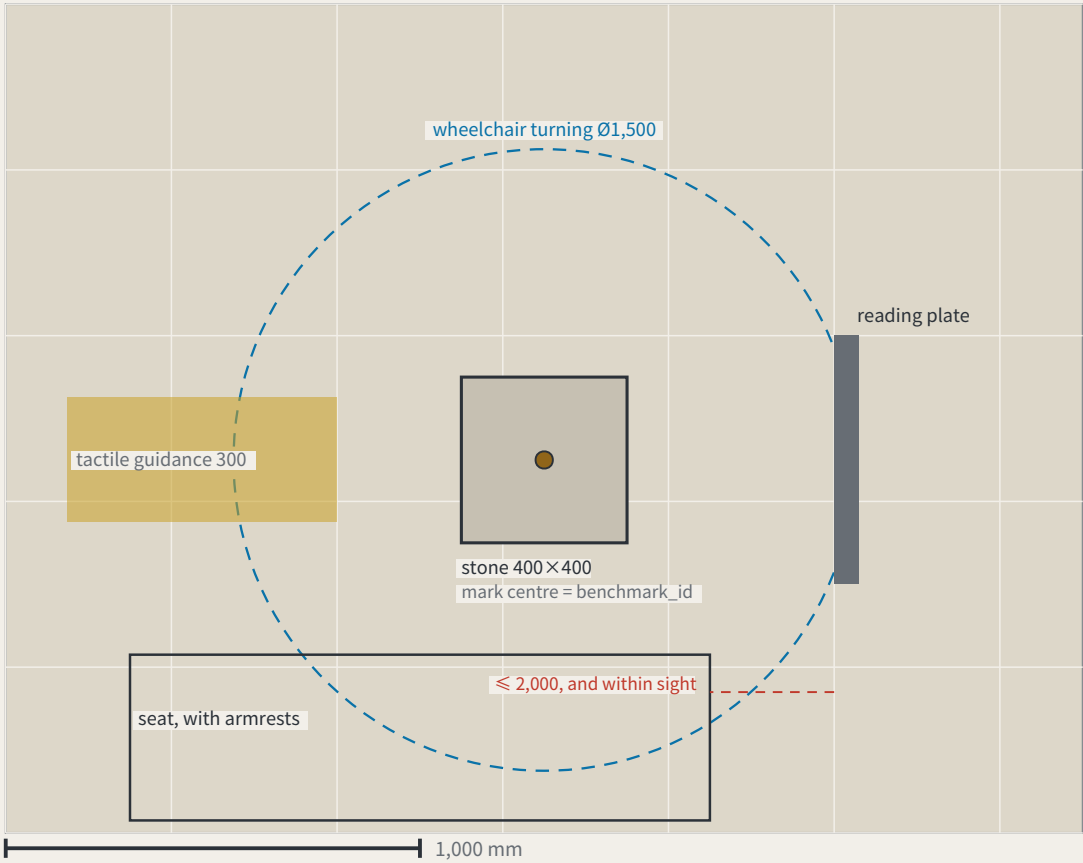
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The benchmark and its reading plate, as a construction detail

FIG.16

a type drawing; it carries no orientation

I. Plan

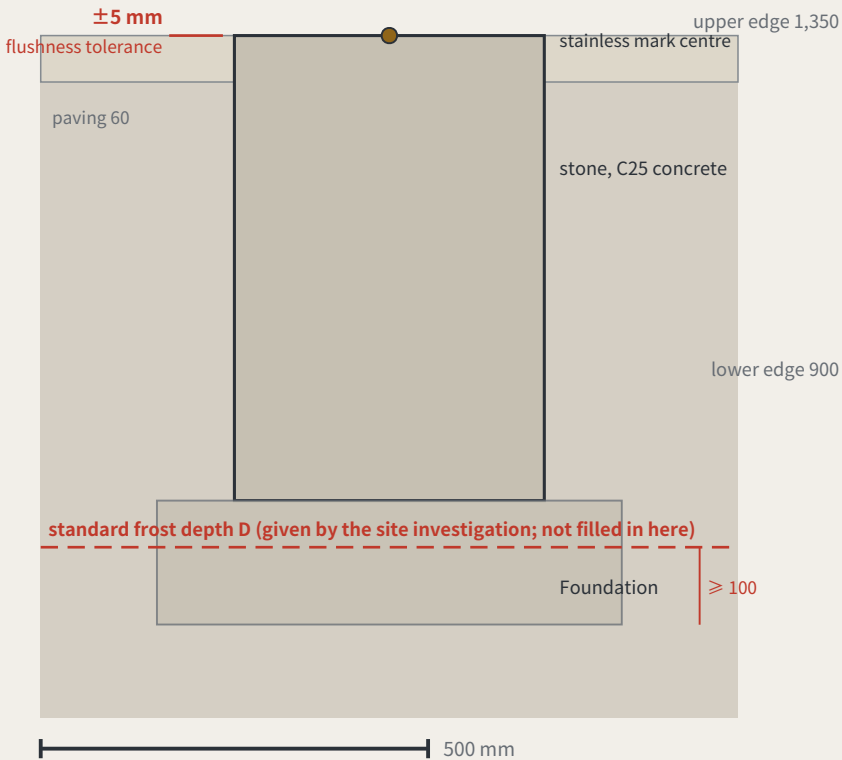


Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

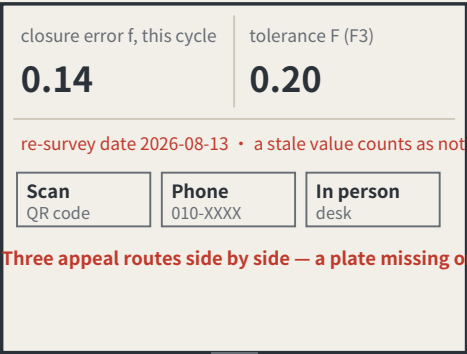
Item	Design intent	Why it is that number	As measured
stone	400 × 400 × 600 mm, C25 concrete, stainless mark centre in the top face	the mark centre is the point a benchmark_id refers to	
Stone top against the paving	≤ ±5 mm	KIT-01 says flush with the ground and no trip hazard; without a tolerance, “flush” is an opinion	
Depth to underside of foundation	≥ the local standard frost depth D + 100 mm	D comes from the site investigation and is not filled in here — this package holds no verifiable frost-depth source for Haidian	
Plate face size	600 × 450 mm, raked 15°	the rake cuts glare and standing water; the face is replaceable without the post	
Plate lower / upper edge	900 mm / 1,350 mm	the band a seated and a standing reader share; P5’ s continuous step-free route should not end at a plate they cannot read	
Appeal panel	QR code, phone number and in-person address, all three side by side	KIT-05: a QR code alone excludes anyone without a smartphone from appealing, which makes persona P4 unworkable	
Tactile guidance strip	300 mm wide, stopping 300 mm short of the stone	the stop line leaves room to stand and take a reading rather than leading onto the stone	
Wheelchair turning space	Ø1,500 mm, headroom ≥ 2,100 mm	the turning space coincides with the reading position, so taking a reading needs no reversing first	
Seat	seat 450 mm high, with armrests, within 2,000 mm of the plate	KIT-03: taking a reading should not require standing and waiting, and the seat must be in sight of the plate	

II. Section



III. Plate elevation (post shown broken)

Specimen values; F is a dimensionless rate, read from scenario_cards.json



the face panel is replaceable without touching the post

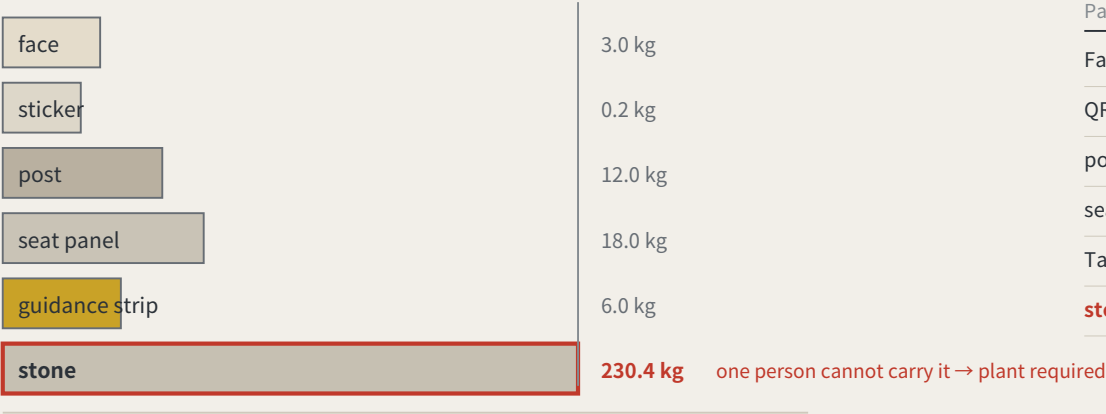
The whole proposal rests on one object, and not one dimension had appeared anywhere in the package — a component library without dimensions is a mood board. Here it is in plan, section and elevation, nine dimensions each with the reason it is that number; read the scale off the bars. The one figure left blank is the frost depth D: no verifiable source here, so the sheet gives the rule and not the number. The as-built column is empty too. Concept design intent; requires detailed design and engineering acceptance.

Maintenance: putting the wear on the parts that can be swapped

FIG.18

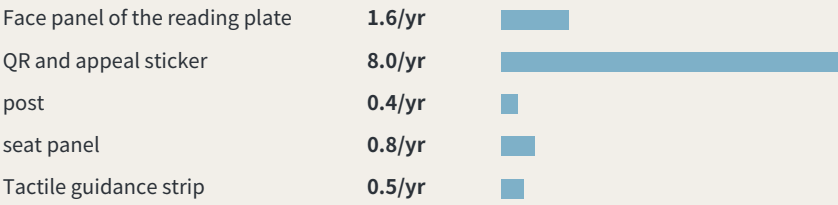
a type drawing; it carries no orientation

I. Order of removal: from what is swapped to what is never moved



IV. What these intervals add up to

Summing the intervals above across 8 benchmarks: about 1.42 replacements per point per year, roughly 11 across the line, about 6 staff-hours a year. Those hours were not in the annual operating table — it counted re-survey sessions only. They are now: paid hours go from 133–214 h to 139–220 h, adding about CNY 680–1,473 a year. Maintenance is not a hidden cost, but it genuinely was not in the table.



Intervals and hours per swap are assumptions, not measurements; the stone is never replaced and is excluded.

II. What fails, how often it is replaced, and whether it touches the datum

Part	How it fails	Replaced	Mass	Tool	Touches datum
Face panel of the reading plate	stale values, scratching, fading	checked each cycle, replaced when damaged	3.0 kg	hex key	no
QR and appeal sticker	the link dies, the surface wears	annual	0.2 kg	none	no
post	impact, corrosion	about 20 years, or after an impact	12.0 kg	socket wrench	no
seat panel	wear, graffiti	about 10 years	18.0 kg	socket wrench	no
Tactile guidance strip	abrasion, and repaving	about 15 years, with the paving	6.0 kg	paving tools	no
stone	impact, excavation, settlement	never replaced	230 kg	plant required	yes

III. After it fails: found → held → replaced → lost

Found	any re-survey that finds the face stale or missing records it in that cycle the plate’ s own rule — a stale value counts as not posted — is the detector
Held	the point is marked returned in datum red, shown at every third-order point in the area bad news belongs where it happened (errata E50)
Replaced	a standard part is swapped without touching the stone or its mark centre one person, one visit; heaviest part 18 kg
Stone lost	the height is re-established from the next order up; the closure record carries the break; the id is retired and never reused reusing an id would quietly join two different points into one history

A monthly re-survey is a maintenance problem first, and this package had said nothing about what fails, how often, or who carries the replacement. One asymmetry organises the sheet: the stone is the only part that must never be replaced and the only one a person cannot carry (about 230 kg) — a benchmark that has moved is not the same benchmark. So the wear goes on the swappable parts. Intervals and masses are estimates, not measured.

Method: trust comes from closure, not from one accurate reading and add one check to the gates.

Leveling is not about measuring accurately; it is about measuring back. Depart from a known point, run the circuit, return — the difference is the closure error. Over tolerance the whole run is void and re-measured, and you may not patch the worst station and keep the rest.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and autonomous shuttles, and AI health, education, legal and daily services. Urban AI governance is the method layer here, not the selling point.

Two prohibitions at the heart of the mechanism

One: amending only the worst station is not allowed. This makes tuning until the metric looks good structurally ineffective.

Two: tolerance may tighten on evidence and may never loosen because a scenario failed. This closes off the moving goalpost.

f is always defined as a deviation, never an attainment score: classification scenarios take 1 minus the consistency ratio, service scenarios the satisfaction range, safety scenarios the sum of false-positive and false-negative rates. All three run the same direction, so $f \leq F$ is always the passing test.

Act One: measure this open call first

228	Merged proposals (git tree, authoritative)
30.3%	Machine-readable disclosure field empty
84 → 8	Distinct strings written → model families

The “machine-readable” disclosure field cannot actually be aggregated: the GPT/Codex family alone is written 44 ways across 103 proposals. The fix is light — split the field into an enumerated family plus free-text detail,

Closure error in the site's own data

412.5 m	Provisional design area to surveyed park
100%	Agreement with the research area

Checking the inferred boundary against OpenStreetMap's surveyed heritage park, two independent routes to the same object do not close. The 43.6 km² level agrees completely; the discrepancy is confined to the 11.4 km² level.

The same measurement also measured this proposal: the submitted spine lies 1,116.7 m from the surveyed park, because it was generated against the provisional boundary. This is stated because a recomputability standard invoked only when it flatters the author is not a standard.

Main front one: low-speed robots and autonomous shuttles

No benchmark, no robot. This turns governance into a spatial design problem: to expand the operating area you must first build measurement points.

Across the field, speed limits, remote and physical emergency stop, on-site safety officers, incident logs, an equivalent non-AI path and scenario-level exit are already the de facto standard. This proposal adopts all of them but presents none as an innovation — they are the admission floor.

The increment is where nobody has been: ice and low-temperature re-survey (zero hits across the field), noise as a number (zero hits), jurisdictional seams, fleet density ceilings, and emergency-access yielding. Their shared structure is that the same system behaves inconsistently across conditions or jurisdictions — which is exactly what closure error measures.

Any safety incident suspends that entire machine type network-wide, not the individual unit. Tolerance scales with kinetic energy: faster or heavier requires a stricter closure clearance first, never an exemption.

Jurisdictional seams: where pilots actually die

8 / 8	Cross-jurisdiction points on this belt
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Crossing jurisdictions is the normal condition, not an edge case. The failure mode at a seam is not a contest over authority but that each side reasonably concludes it is not theirs, so the device keeps running unreviewed until something happens. The rule: if neither side holds a valid reading, the section is closed — so the default consequence of inaction is that the device stops.

Main front two: AI public services

Do not measure the average; measure the dispersion. The risk is not in mean accuracy but in how much the conclusion differs when different people ask the same question — which is precisely what closure error measures.

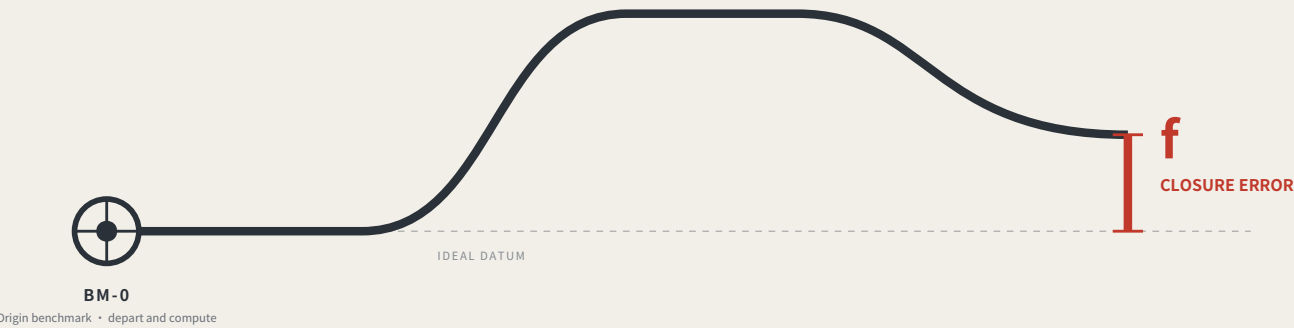
Three non-waivable boundaries: prescriptive judgements must be made by qualified people and logged; the equivalent non-AI path is permanent and must not be slower; and no profiles of identifiable individuals and no cross-scenario linkage.

Statement of boundaries

Everything here is open collaborative concept advice on provisional substitute boundaries. It does not replace statutory planning and constitutes no government determination, approval basis or implementation commitment. No floor area ratio, building height, density, setback or road redline figures are given, and no heritage protection zone or blue line is inferred. Responsibility is written as role types only, all subject to negotiation. Final judgement rests with people and professional teams.

THE LEVELING LINE

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

jiangmuran • Open call for the Centennial Jing-Zhang AI Innovation Belt • Concept advice on provisional boundaries; not a substitute for statutory planning

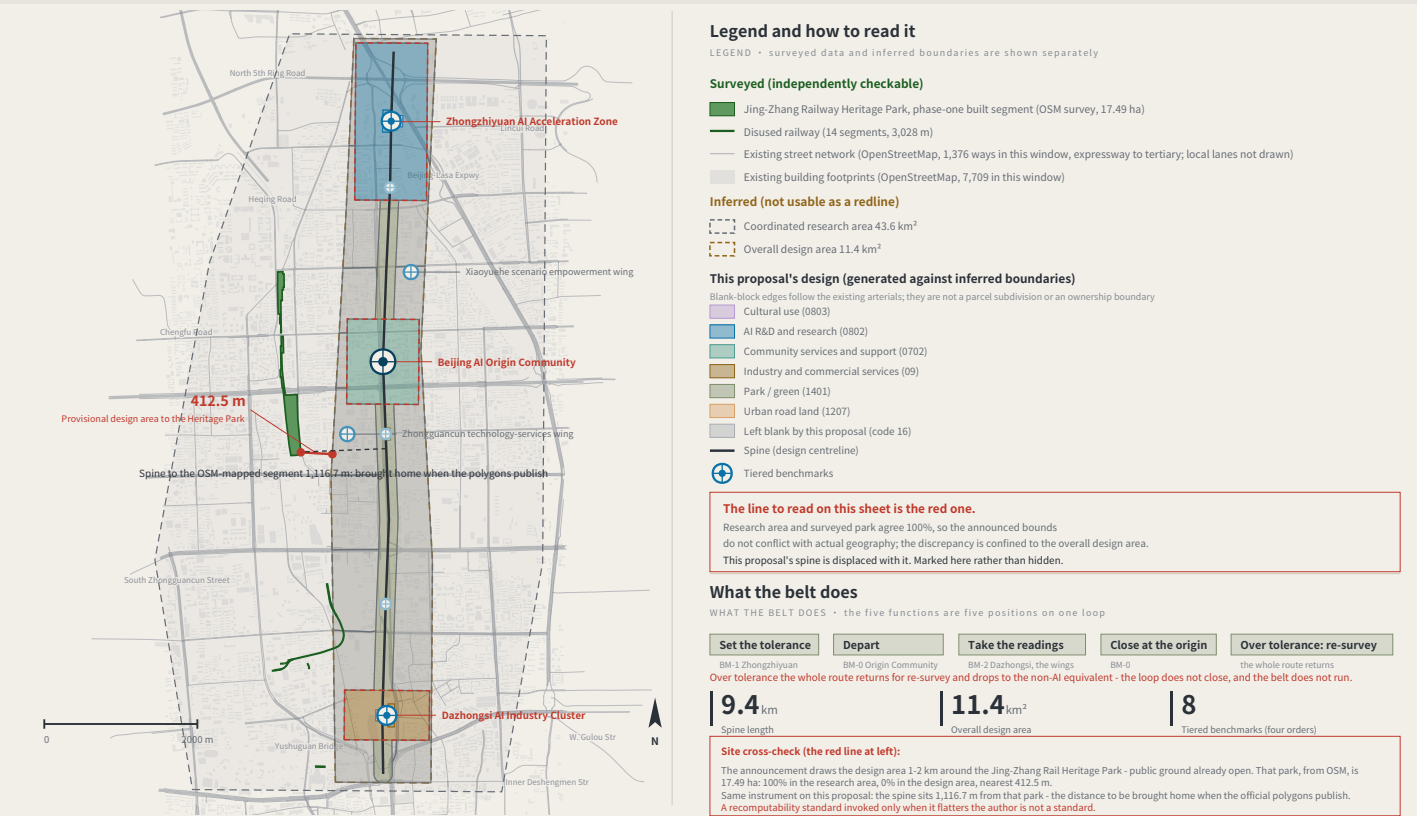
FIG.00

OVERALL CONCEPT AND SITE OVERVIEW

FIG.01

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Surveyed data © OpenStreetMap contributors • ODbL 1.0 • Overpass API 2026-08-08. Script and conventions in visual/assets/osm_reference.json; re-runnable. OSM is crowd-sourced and the park polygon may cover only the mapped section; the provisional boundary is itself inferred. This sheet reports the difference and does not adjudicate which is right.

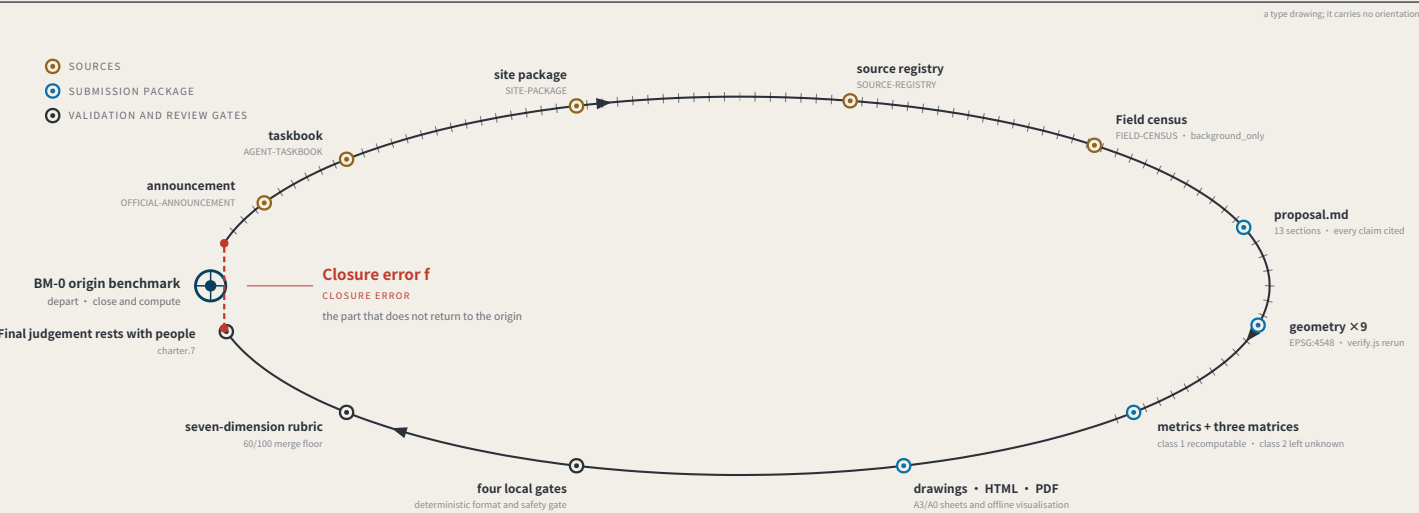
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Data to 2026-08-19 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

EVIDENCE CHAIN AS AN UNCLOSED LEVELING CIRCUIT

FIG.02



Act One: measuring this open call itself

MEASURING THE OPEN CALL ITSELF • as of 2026-08-13 • 793 merged • 793/793 fetched • PR numbers past 1000

793

Merged proposals (git tree, authoritative)

The gallery index lists 507 at the same moment — 287 behind

18.2%

Machine-readable disclosure field empty

144 of 793 still hold the scaffold placeholder or are empty

228 → 9

Distinct strings written → actual model families

The GPT/Codex family alone is written 85 ways across 355 proposals

The "machine-readable" disclosure field is not machine-readable.

charter.6 requires disclosing the generation method and charter.5 requires it structured and agent-readable, yet none of the four gates checks this field. Nobody can aggregate "which models produced this call" from it.

The fix is light: split "model" into an enumerated family plus free-text detail, and add one enumeration check.

This proposal is the instrument for the missing last step: compute the closure error, and give it consequences.

Registers answer "what did the system declare" and assessments answer "what do experts think"; closure error answers "how much does the conclusion differ across nodes and across time for the same public question".

Sources: visual/assets/census.json, field_map.json and the first-round (184-file) snapshot agent_declarations.json, all shipped with the package. The authoritative source is the GitHub git tree, not the gallery index, which lags. Generation scripts are in the accompanying issue and are re-runnable. The corpus grows daily; recompute before citing.

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Data to 2026-08-19 (sources.json; 1 not obtained)

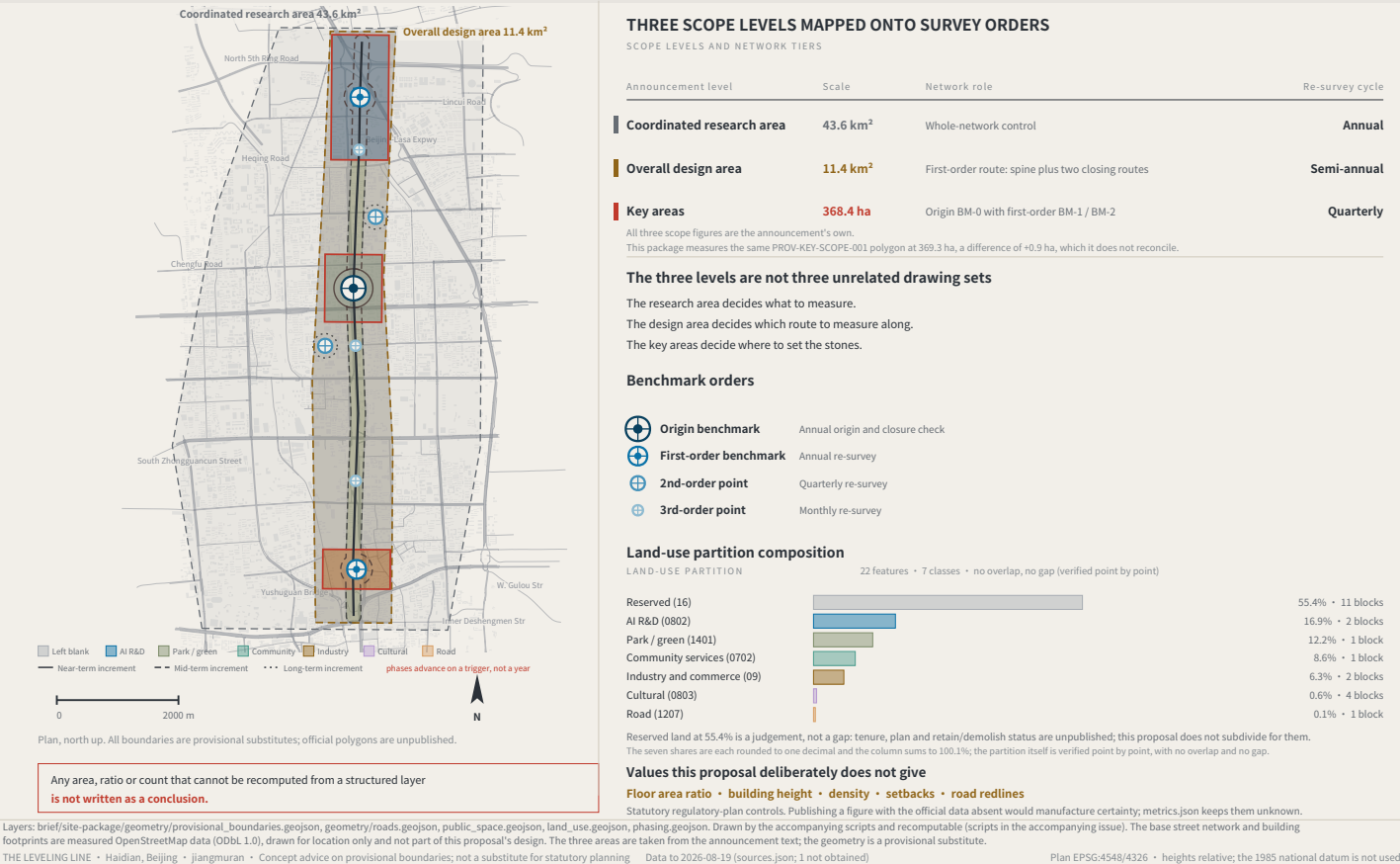
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

LAND-USE STRUCTURE AND THE THREE SCOPE LEVELS

FIG.03

7 land-use classes, no overlap and no gap, verified point by point; the 55.4% left blank is a judgement — tenure and control plans are unpublished.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



a type drawing; it carries no orientation

The chain of custody for innovation elements

ELEMENTS AND THEIR CHAIN OF CUSTODY

Every element answers the same three questions: which space carries it, which benchmark holds its reading, and what gets re-measured.

Element		Spatial carrier		Benchmark		Re-survey item
Land and space	→	Key-area renewal parcels	→	BM-1 / BM-0 / BM-2	→	Delivery rate of promised space
Compute	→	Consolidated compute nodes	→	BM-1	→	Share of public compute open to application
Data	→	Data-factor circulation venue	→	BM-2	→	Authorisation-chain completeness
Scenarios	→	Real-use venues in the two wings	→	BM-21 / BM-22	→	Enforceability of scenario exit
Funding	→	Zhongguancun technology-services wing	→	BM-21	→	Deliberately blank – see the note below
Talent	→	Talent district and third places	→	BM-0	→	Re-survey of service satisfaction

The funding row has no re-survey item. That is a judgement, not an omission.

Investment arrangements are decisions for market actors; an urban design proposal must not turn them into governance metrics, still less into fiscal commitments.

Writing what is not yours to govern into a metric system is its own kind of overreach.

Six global cases, one question

SIX CASES, ONE QUESTION: WHAT MAKES ITS TRUST RE-MEASURABLE?

Numbering	Case	Trust mechanism	Re-measurable	Transferable point
C1	Algorithm registers	Public register of purpose, data, owner and appeal route	High	The information base for a benchmark plaque
C2	Risk-tiered AI legislation	Duties differentiated by risk class	Medium	Supports the tiered setting of tolerance F
C3	Standardised testing frameworks	Comparable reports from a common toolkit	High	Gives re-survey its technical form
C4	Algorithmic impact assessment practice	Mandatory ex-ante and ex-post review for high-risk uses	Red-high	Supports the strictest tolerance for health scenarios
C5	Civic data-stewardship practice	Data use decided by a citizen agenda	Medium	Supports public re-survey rights at third-order points
C6	Open-source reproducibility norms	Conclusions must ship a re-runnable artifact	High	This package ships a runnable verifier

All six point at the same gap: they register and they assess, but none institutionalises returning to the origin and computing.

A register says what a system declared; an assessment says what experts think. Neither answers how much the conclusion differs when the same public question passes through different nodes at different times.

Case material is drawn from public institutional documents and public reporting. Mechanisms only are cited: no text or images copied; no marks used; no non-public data; and no fabricated company lists; investment figures or output values. Element-to-space correspondence is in geometry/land_use.geojson and public_space.geojson.

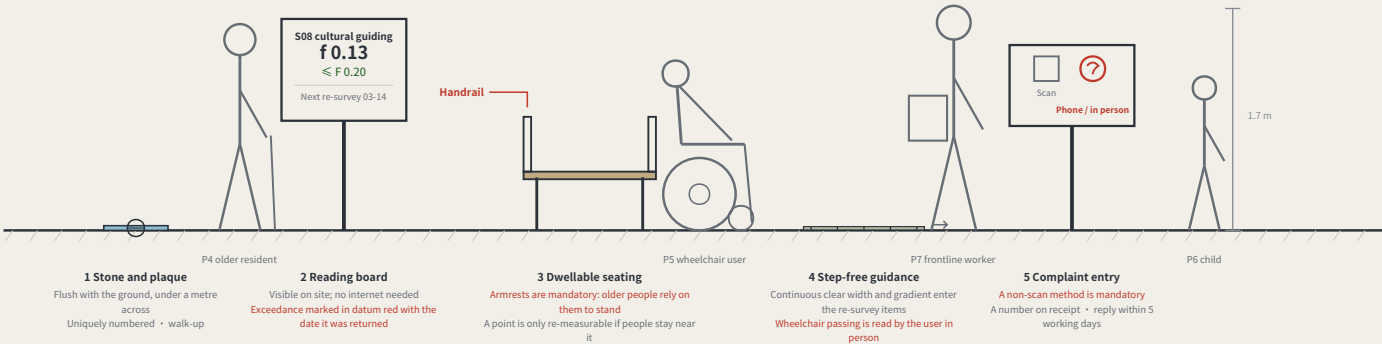
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Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

a type drawing; it carries no orientation

What a benchmark looks like at eye level

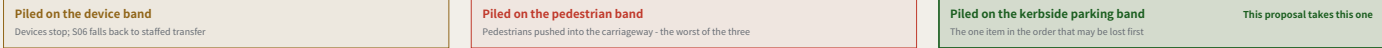
A BENCHMARK AT EYE LEVEL • third-order BM-301 • elevation 1 m = 138 units



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

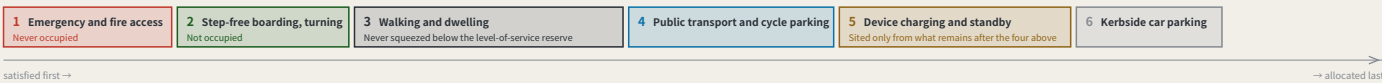
The same kerb, after snow clearance

In winter the order becomes one real trade-off



Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER • sequence and precedence only, no widths



The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. The plate's f0.13 is an illustration, not a reading: none is measured yet. Component specifications are in geometry/public_space.geojson and the scenario cards; positions await official boundaries and title verification.

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Three AI pilgrimage landmarks

THREE LANDMARKS • spoken in engineering language; no spectacle

L1 The origin benchmark stone

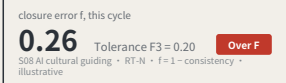
BM-0 • AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

L2 The closure stele

BM-1 • Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. The value of this landmark is that it is allowed to look bad.

L3 return-to-datum point

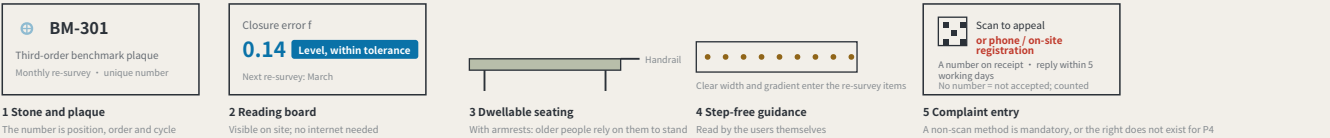
BM-2 • Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

Kit of parts for public space: five standard components

KIT OF PARTS • common specifications and open drawings, so any new node joins in one language



Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

BM-0	Origin benchmark	Annual origin and closure computation
BM-1 / BM-2	First-order benchmark	Annual re-survey
BM-2x	2nd-order point	Quarterly re-survey
BM-3xx	3rd-order point	Monthly re-survey
RT-N / RT-S	Closing route	Semi-annual re-survey of the whole line

Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

Monthly	Community day	third order BM-301/BM-302/BM-303	P4 • P5 • P7
Quarterly	Scenario open day	second order BM-21/BM-22	P2 • P3
Semi-annual	Route re-survey	first order, routes RT-N / RT-S	P1 • P2
Annual	Zeroing ceremony	L3 return-to-datum point	all four review parties present

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate international communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. F is always a deviation: larger is worse, and f ≤ F is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

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Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

a type drawing; it carries no orientation

An executable form of coordination: how two levelling networks join



A joint observation produces a number, not an intention:

Each network reads the same public question set; the difference is the cross-network closure error. Within a jointly agreed tolerance, the two networks' records may be mutually recognised; over it, each re-surveys, and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

Partner	Stage	Exchanges	Explicitly does not exchange	Precondition for the interface
BDA (Yizhuang)	Volume production and high-level AI road testing	Mutual recognition of closure records across complementary speed domains	Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds	Both parties confirm the speed-domain split and test protocols
Future Science City	Corporate research institutes and engineering	Scenario demand and real user density	No claim on IP; no scheduling on anyone's behalf	Wing benchmarks built and a first baseline taken
Huairou Science City	Large facilities and basic research	The measurement method itself: how f is defined, cycle length, the statistics behind revising F	No scenario data is exchanged; the coordination is methodological, not applied	The closure convention published and open to methodological review
Beiwei Community	The innovation community the taskbook names	Reciprocal joint points; exchanged review parties	This proposal does not know its positioning and does not set its benchmark order	Each party names at least one publicly reachable joint point
Beijing-Tianjin-Hebei	Cross-provincial coordination	Cross-provincial recognition of the tolerance convention	No direct recognition of records; without a shared convention they are not comparable	Tolerance definition and revision procedure agreed across provinces

None of the five rows has been confirmed by the party named.

No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

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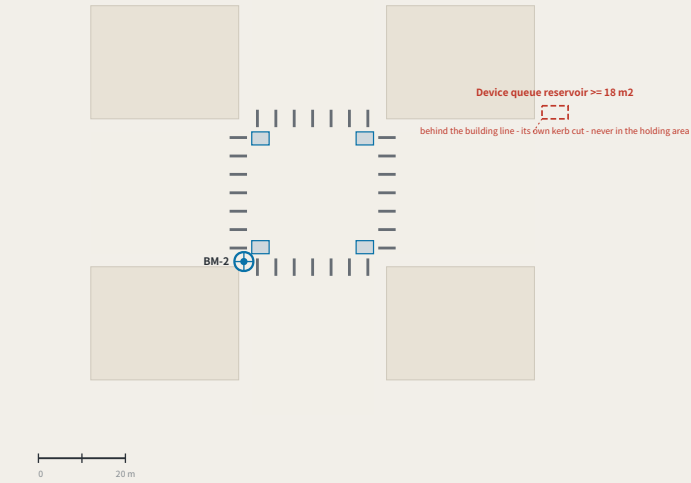
Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

Four-quadrant pedestrian connection and device queue reservoir at Dazhongsi

FIG.12

I. Four-quadrant plan

FOUR-QUADRANT PLAN • TYPE APPLIED AT BM-2, NOT A SURVEY



2. Dimensions: design intent / as-measured

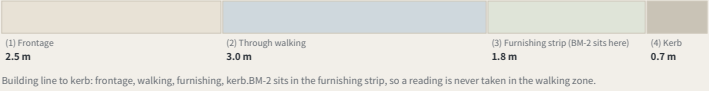
DIMENSIONS: DESIGN INTENT vs AS-MEASURED

Item	Design intent	As measured
Pedestrian holding area, per quadrant width >= 3.0 m x depth >= 4.0 m	>= 12 m²	
Crossing effective clear width after deducting poles, covers and the furnishing strip	>= 4.0 m	
Kerb ramp gradient Barrier-Free Environment Construction Law and current codes	<= 1:12	
device queue reservoir set behind the building line, with its own kerb cut	>= 18 m²	
Reservoir to holding area, clear distance must not cross any desire line	>= 2.0 m	
BM-2 stone to pedestrian edge the stone is flush with the ground and is not a trip hazard	>= 0.6 m	

The right-hand column is deliberately empty.
Capacity is computed on site from measured clear width. A number filled in without measuring is manufactured certainty. Completed before any device is admitted.

3. Kerb allocation section (order and intended widths only)

KERB ALLOCATION SECTION



This is a type drawing applied at BM-2, not a survey of the existing junction. Every dimension is design intent and the as-measured column is deliberately empty: carrying capacity must be computed on site from measured effective clear width. Junction geometry, channelisation and signals rest with the road authority and a traffic study.
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The three key areas in section, at one scale

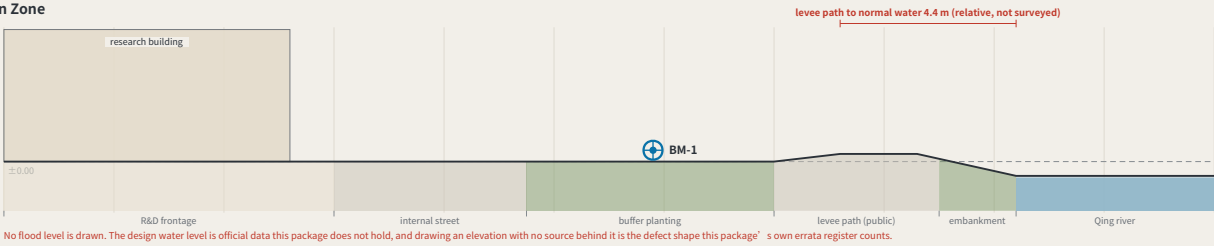
FIG.13

Zhongzhiyuan AI Acceleration Zone

±0.00 = BM-1

192.9 ha • BM-1
dominant use R&D 100%
2 land-use classes inside

External arrival. The area is 100% single-use, so public access to the Qing river runs through the block or not at all — what this section sets out is a public route across the R&D frontage and onto the levee path.

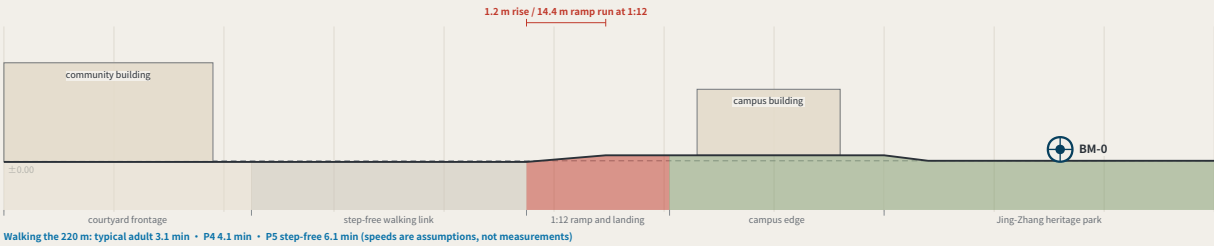


Beijing AI Origin Community

±0.00 = BM-0

104.3 ha • BM-0
dominant use community services 94%
2 land-use classes inside

A 1.2 m grade change, and step-free continuity. The campus edge ends in steps, and P5 needs a continuous step-free route. The reason this section exists is to put a 1:12 ramp with a landing where the steps are.

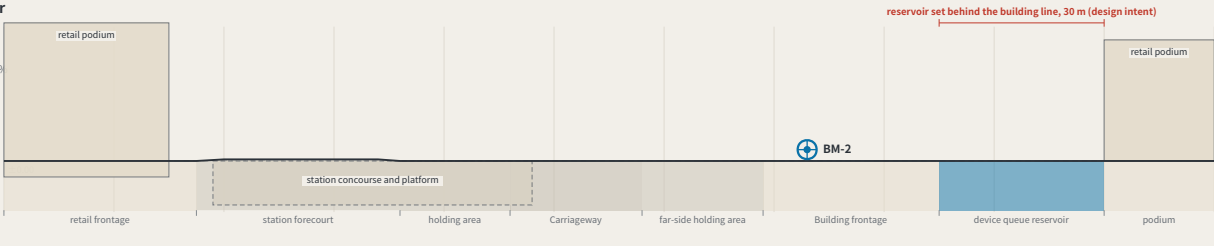


Dazhongsi AI Industry Cluster

±0.00 = BM-2

72.0 ha • BM-2
dominant use industry and commerce 99%
2 land-use classes inside

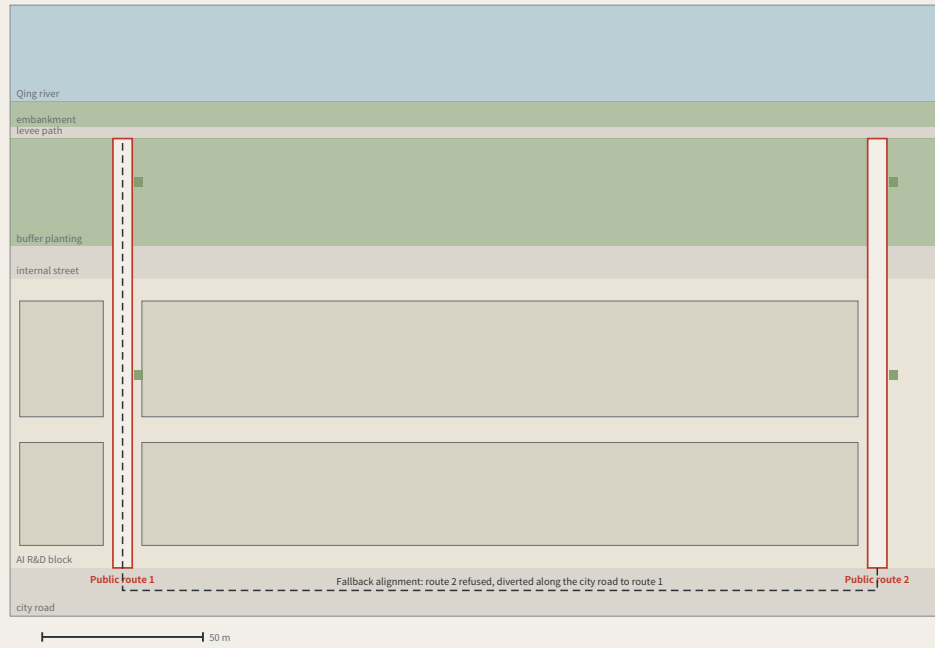
A station below grade and a queue behind the building line. The only section here with anything below ground, and the only one with a carriageway. The device queue reservoir sits behind the building line with its own kerb cut, clear of the crossing desire lines — the four-quadrant plan is FIG.12.



One horizontal scale, one datum convention, no vertical exaggeration. They are placed together so the charge that the three are one template used three times can be tested: one reaches a river across a 100% single-use block, one is step-free continuity over a 1.2 m rise, one is a station below grade with the device queue behind the building line. ±0.00 is each area's own benchmark — absolute elevations need official levelling data this package does not hold, so they are not drawn. Concept design intent, not surveyed profiles.
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The through-block public route to the Qing river

FIG.14



Dimensions: design intent and as built

The right column is deliberately empty — effective clear width is filled in on site

Item	Design intent	As measured
Public route clear width two abreast, and a wheelchair passing	>= 6.0 m	
Route centre spacing a 939.0 m frontage needs 4 routes, 234.8 m apart	<= 250 m	
Longitudinal gradient 133.5 m rising 1.4 m, an average of 1:95, step-free throughout	no stretch steeper than 1:20	
Opening hours a route that can be locked is not public access; it is permission	24 h, ungated	
Spacing of places to sit standard component KIT-03, with armrests	<= 60 m	
Levee path clear width two-way walking, passing	>= 3.5 m	
Building setback from the route the ground floor may not face it with a solid wall	>= 3.0 m	

The count is not a preference. It is a division.

Under the provisional substitute boundary the northern frontage of Zhongzhiyuan runs 939.0 m. At a maximum centre spacing of 250 m that is 4 public routes, 234.8 m apart. Replace the boundary with an official polygon and the number moves with it — it is computed from the geometry at build time, not written onto the sheet.

What one gate costs

With the route open it is 133.5 m from the city road to the levee path. With route 2 refused and the walk diverted along the city road to route 1, it is 368.3 m. For P5, walking at 0.6 m/s, 3.7 minutes becomes 10.2 minutes. That is the working unit of a public route that can be locked.

The route along its length: city road to levee path

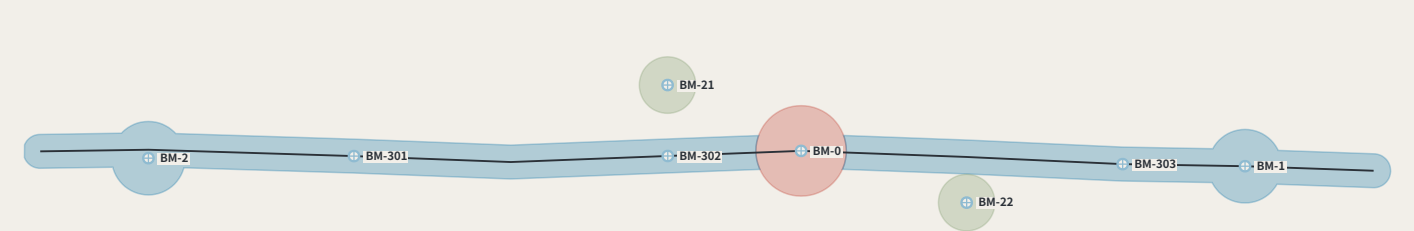
2.6× the scale of the plan • no vertical exaggeration • 133.5 m rising 1.4 m, an average of 1:95



FIG.13's section says this edge needs a public route across the R&D block onto the levee path, and no drawing showed it. This is that route: 6.0 m clear, centres at most 250 m apart (4 of them across a 939.0 m frontage), open 24 hours and ungated. The fallback alignment and the cost of a refusal are drawn. A type drawing, not a survey; dimensions are design intent, the as-measured column is empty; boundaries are provisional.
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Phasing: advanced by closure results, not by dates

FIG.15



Phases advance on closure results, not on years

Phase	Increment	Spine added	Benchmarks this phase adds	Condition that opens it (not a date)	Re-surveys/yr	Annual running cost
Near	32.1 ha	0.64 km	BM-0	none — no prior closure needed	1	CNY 2,484–7,922
Mid	234.8 ha	8.80 km	BM-1, BM-2, BM-301, BM-302, BM-303	four near-term items done; f <= F × 2	39	CNY 13,864–45,492
Long	25.1 ha	—	BM-21, BM-22	mid phase: f <= F × 2	47	CNY 19,444–63,922

The table above phases the eight points that exist and their cumulative cost. FIG.21 shows that meeting the fifteen-minute rule needs 9 more third-order points (a further CNY 22,005–76,358 a year), falling by order into the mid and long phases — this is not the phasing of a network that already complies.

The first phase is a stone, a plate and one reading a year

The near-term increment is 32.1 ha and holds BM-0 alone: 1 re-survey a year, CNY 2,484–7,922 to run. That line is the answer to "this mechanism is too heavy" — it can start from one point, and that point costs what a sub-district office can approve by itself.

The condition is a number, not a year

The mid phase does not begin because a second year arrived. It begins because the four near-term items are complete and f <= F for two consecutive cycles; the long phase the same. Once may be luck — the same rule this proposal writes at PATH-4 of the conversion path, applied to phasing. Fail it and the programme stays where it is, which makes the phasing table an exit table as well.

Which four are the four near-term items

The condition that opens the mid phase reads "the four near-term items complete" and nothing in the package says which four — and the proposal contains two different sets of four: the near-term projects R1–R4, and the ones needing no official data, R1–R3 and R8. The second set contains a long-term item, so reading the gate against it makes the gate circular. Taking the first: R1 datum stone L1 and the public evidence hall — one complete closure and its reading published within the first cycle R2 first public setting of the tolerance F — a first version of F1/F2/F3 each, posted R3 the four-week closure trial on cultural guidance S08 — closure recomputed and the consistency ratio published within four weeks R4 third-order benchmarks set out at communities and rail stations — one recorded reading at each of the three community points

Two naming errors this sheet found

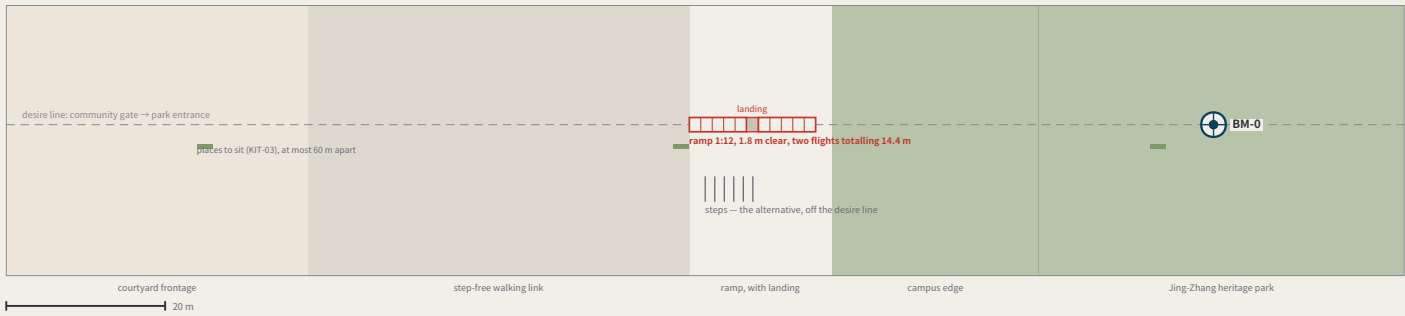
Dropping each benchmark into the phase polygons contradicts two of the phase names. The mid phase was named for "three first-order points" and contains two (BM-1, BM-2); the third is the datum BM-0, which is in the near phase. The long phase was named for "the full three-order network", and the three third-order community points already fall inside the mid phase — what the long phase actually adds is the two second-order wing access points. The names were written for the intent and never checked against the polygons. Both are corrected in the layer, and every cell of this table is computed from the geometry rather than typed.

phasing.geojson has shipped for a long time, been repaired once and enters the recompute path, and no drawing had ever shown it. This sheet draws the three increments and costs each phase's year with the same model as build_operations, not a second set of coefficients. Phases open on closure results, not on years. The geometry is concept advice on provisional boundaries and must be recomputed when official ones are published.
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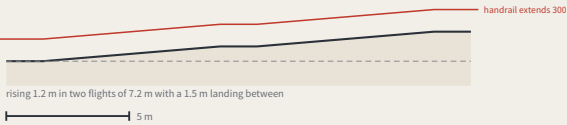
The step-free link from the origin community to the heritage park

FIG.17

I. Plan



II. The ramp along its length (no vertical exaggeration)



What offsetting the ramp would cost

The usual arrangement puts the steps on the desire line and offsets the ramp. Offset it by 22 m and whoever uses it walks 44 m further every time — out and back. For P5, at 0.6 m/s, that is 1.2 minutes a trip; at twice a week for a year, 4.6 km. Nobody on the steps walks a metre further for the ramp being on the line: steps are shorter, so they sit beside it.

Walking the 176 m

typical adult	2.4 min	
P4	3.3 min	
P5	4.9 min	

speeds are assumptions, not measurements

Dimensions: design intent and as built

The right column is deliberately empty — as-built values are measured on site

Item	Design intent	Why it is that number	As measured
Ramp gradient	1:12	step-free throughout; the ramp is on the desire line and the steps are its alternative	
Ramp clear width	≥ 1.8 m	two people passing, or a wheelchair with someone alongside	
landing	one every 750 mm of rise, at least 1.5 m long	a 1.2 m rise split into two flights — there has to be somewhere to stop	
Handrail	both sides, extending at least 300 mm past each end	a handrail that stops at the top of the slope stops where it is still needed	
Level approach at each end	at least 1.5 m level	level ground before and after, so nobody has to turn on the slope	
Spacing of places to sit	≤ 60 m	standard component KIT-03, 450 mm high with armrests	
Steps	beside the ramp, off the desire line	anyone who prefers steps is not sent around — steps are shorter than the ramp	

FIG.13' s section says this place exists to put a 1:12 ramp where the steps are, and no drawing in the package had shown it. One design decision: the ramp sits on the desire line and the steps beside it — offset the ramp by 22 m and its user walks 44 m further every trip, while nobody on the steps walks a metre more. A type drawing, not a survey; dimensions are design intent and the as-built column is empty.

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Monuments beside heritage fabric: the no-drilling construction and its setback rule

FIG.19

I. Choosing the construction by distance from the fabric



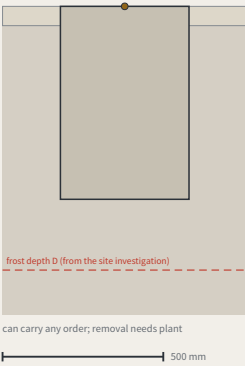
d = horizontal distance to the outer edge of the fabric

The rule is complete; its input is missing

Which line d is measured from depends on the officially published heritage protection area and its construction-control zone. This package does not hold that layer: each of the 3 shipped constraint features records it as a data gap in its own note, and this proposal does not infer it. So the sheet gives the rule, the thresholds and the three constructions, and leaves that line to whoever publishes it — guessing a protection boundary to make the drawing look complete would mean inventing the one thing here that is not this proposal' s to decide.

II. Two constructions, one scale

Buried: foundation below the frost line



Surface-set: bonded, no drilling, no excavation



III. What each construction costs

Construction	Applies when	Foundation	Orders it may carry	Re-survey cycle	What removal leaves
Buried stone	d ≥ 5 m	below the standard frost depth D + 100 mm	datum / first / second / third	per order	plant to lift out; leaves an excavation
set-back independent foundation	2 ≤ d < 5 m	below frost line, set back ≥ 2 m, unanchored	second / third	per order	plant to lift out; leaves an excavation
Surface-set marker (no penetration)	d < 2 m	no foundation; bonded to existing paving, no drilling	third order only	monthly, and tied to the nearest buried point each time	lifting it leaves only a bond mark, which cleans off

The call is for a belt along a hundred-year-old railway; this proposal' s central act is sinking concrete. Three constructions, chosen by distance from the fabric, with the cost stated: without excavation there is no foundation below the frost line, so the surface-set marker may not carry first- or second-order duty. Where d is measured from depends on the official heritage layer, absent here and recorded as a gap in constraints.geojson.

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Reading after dark, without lighting the benchmark

FIG.20

I. Geometry: where the light comes from, and returns to



II. Three approaches, compared by how they fail

Approach	Needs power	How it fails	Light trespass
Retroreflective face, reader brings the light	no	the face gets dirty → visible in daylight → caught before it stops being readable	none
Low-level floodlight	yes, at every point	the lamp dies → the plate looks intact and cannot be read	yes, needs controlling
Internally lit face	yes, at every point	the lamp dies → the whole face goes, and it still looks like a plate	yes, and aimed at the reader

The first is chosen not because it is cheap but because its failure is visible. When a lamp dies the plate still looks intact and can no longer be read — exactly the thing this proposal exists to prevent, a measurement whose failure cannot be seen. A dirty retroreflective face is simply dirty, visible in daylight to anyone walking past, well before it stops being readable.

plate face, raked 15°
retroreflective film, returns light to its source
a specular surface would bounce it straight into the reader — hence not specular

III. What is not lit

No fixed luminaire	the reader brings the light; no power supply, no luminaire to maintain
The stone is not lit	it sits flush with the ground; lighting it adds no legibility, only night-time glare
Nothing aimed upward	no supplementary light may be emitted above the horizontal
Nothing crossing the boundary	the added vertical illuminance at neighbouring dwelling windows must meet the local limit — that figure comes from the local standard, which this package does not hold

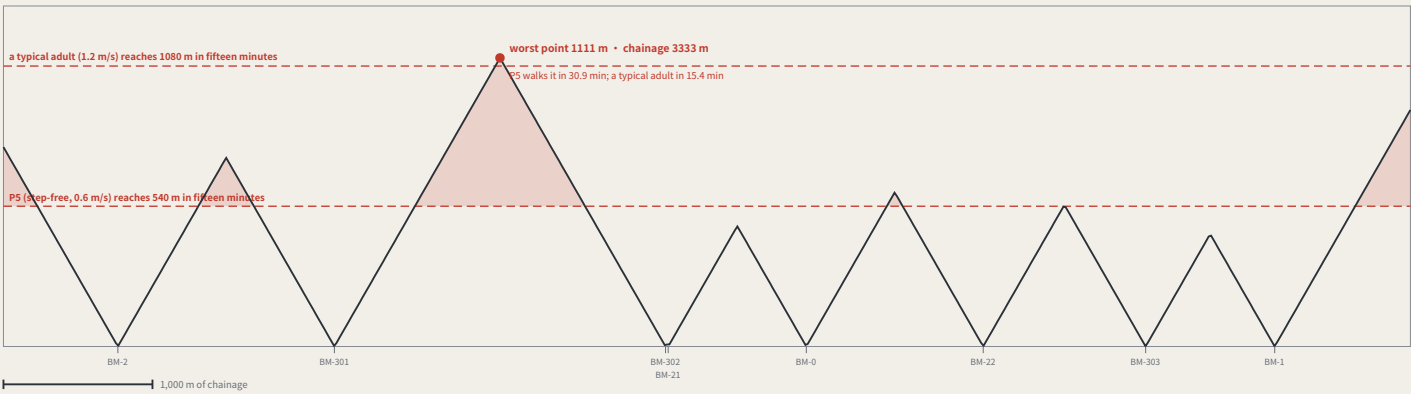
Three third-order points are read monthly by residents, and much of the year those days fall after dark — and the package had never said how the plate is read at night. The obvious answer is a lamp, and it is the wrong one: it needs power, needs maintaining, throws light into the windows opposite, and when it dies the plate looks intact and cannot be read — a measurement whose failure is invisible, which is the thing this proposal exists to prevent. So the face is retroreflective with no fixed luminaire and the reader brings the light; a dirty face is simply dirty, in daylight.

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How far the nearest benchmark actually is: this proposal's own rule, applied to itself

FIG.21

I. Distance to the nearest benchmark, along the spine



II. Turning that sentence into a spacing rule that can be checked

Where the rule comes from	this proposal' s own repeated claim: a right of review you must walk fifteen minutes to reach has not been given
Interior spacing	chainage between adjacent points ≤ 2 × 540 m = 1080 m
Both ends of the line	end of line to its nearest point ≤ 540 m
Which speed the rule uses	P5' s 0.6 m/s, not a typical adult' s 1.2 m/s — setting the rule by the faster figure sets it at a standard most people cannot use

III. How far short the current layout falls

6 of the 9 segments between the 8 existing points fail. Closing the gap needs 9 more third-order points (or 3 if the rule is only held to an average walker). They are not free: third-order points are read monthly, and every one of them enters the annual operating table. This sheet does not place them — putting 9 coordinates on a provisional substitute spine is the manufactured certainty this register counts. The rule ships; the positions wait for the official alignment.

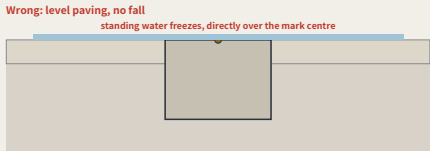
This proposal argues repeatedly that a right of review you must walk fifteen minutes to reach has not been given, and its entire participation argument rests on anyone being able to walk up to a benchmark and take a reading — and nobody had ever measured that walk. Measured: the worst place on the line is 1111 m from the nearest benchmark, 30.9 minutes for P5, twice the limit this proposal holds others to. The rule ships as a spacing requirement; the positions wait for the official alignment.

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Winter: water, ice, and two of this package's rules in collision

FIG.22

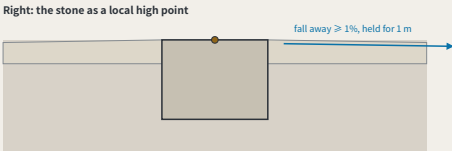
I. Two ways of doing it, at one scale



Level paving with no fall: compliant in September, the only ice in the sidewalk in January

1,000 mm

The fall is drawn true: 1% over 1 m is 10 mm, which is barely visible at this scale — and a slope nobody can see on a drawing is a slope nobody checks on site, while every dimension on the sheet is still satisfied.



The stone is the local high point: flush with the paving beside it, and that paving falls away

II. What that requires of the construction

Item	Requirement	Why
Stone top against the paving	flush, tolerance ± 5 mm	KIT-01: no trip hazard
Fall around it	$\geq 1\%$, away from the stone, held for ≥ 1 m	makes the stone a local high point — water leaves instead of gathering
Protective ring	none	a recessed ring is a bowl; a raised one is the trip hazard KIT-01 forbids
Surfacing	permeable or tight-jointed; no joint channel running inward	a joint channel leads water to the mark centre
Winter reading	taken after clearing, with the condition recorded	a cleared reading and an uncleared one are not directly comparable

III. Three things a winter reading adds

Clear it	clear snow and ice from the stone and 300 mm around it before reading a reading taken over a film of ice is measuring the ice
Check the drainage	confirm the stone is still the high point: no ponding marks or ice within 1 m a ponding mark is evidence the drainage has already failed, and it appears before the ice
Record the condition	mark the cycle record as a winter condition, and whether the stone was cleared cleared and uncleared readings are not directly comparable — unrecorded, nobody knows whether they should be

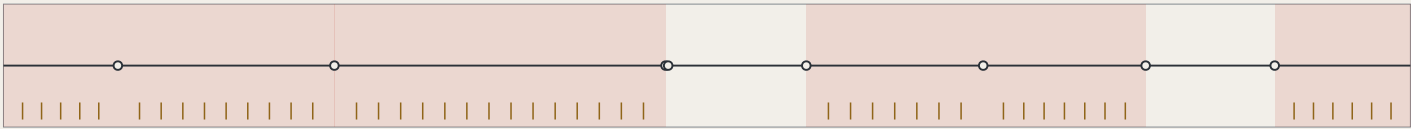
The three third-order points are read monthly, January included, and this package had never said what that costs. KIT-01 requires the stone flush with no trip hazard and FIG.16 gave that ± 5 mm — but a mark flush in level paving sits in standing water: the tolerance that makes it safe in September makes it dangerous in January. The answer is a local high point, not a ring.

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Finding the nearest benchmark: which way, and how far

FIG.23

1. Wayfinding only in the segments where this proposal is already in breach



Red marks the segments FIG.21 judged to fail the fifteen-minute rule; one wayfinding post every 150 m, 48 in all. Compliant segments get none — putting up a sign where it is already close enough is decoration.

1,000 m

2. The four things the plate must answer

← BM-302 410 m • P5 approx. 11 min	Direction ← BM-302 / BM-0 → Either end of the line may be nearer; only at the midpoint are both sides equally far
BM-0 → 520 m • P5 approx. 14 min Standard part KIT-04, no new structures	Distance 410 m / 520 m metres, because metres can be checked by the person who walks them
	Time P5 approx. 11 min / 14 min At 0.6 m/s, the same figure as in every other drawing in this package (personas.json)
	Numbering BM-302 • BM-0 A reading taken on arrival is attributed by it; no need to ask afterwards where the person was standing

3. What wayfinding cannot do

These 48 wayfinding posts will not make 1111 m any shorter. What they do is let a person know before setting out how far it is, which way, and what the point is called on arrival. This is mitigation, not a fix. The fix is the 9 third-order points in FIG.21, which give no position until the official alignment is published. Writing the wayfinding up as the solution is exactly the kind of substitution this package has kept catching in itself — using one small thing it can do to cover one large thing it cannot.

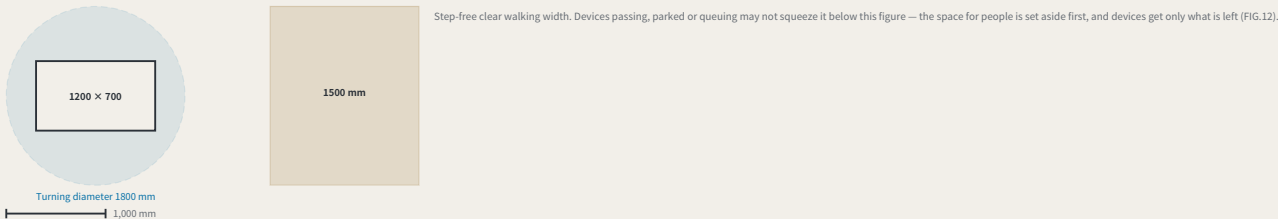
FIG.21 measures the farthest point on the whole line at 1111 m from the nearest benchmark: 9 segments, of which 6 fail this proposal's own fifteen-minute rule. Closing the gap takes 9 third-order points, but their positions await the official alignment. People still have to find the points that already exist. This drawing supplies wayfinding: placed only inside the segments already in breach, one every 150 m, 48 in total, using the existing standard part KIT-04, adding no new structures. Wayfinding will not make 1111 m any closer — this is mitigation, not repair.

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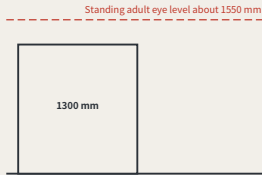
The device envelope: what exactly goes in those 18 m²

FIG.24

1. Envelope and turning, plan



2. Elevation: why the height is this number



Height sits below eye level so queued devices do not cut the sightline across the sidewalk. No external basis; set by this proposal — eye height varies from person to person, and the sheet gives the reasoning, not a clause to copy.

3. The envelope items, and the constraint source of each

Item	Value	Constraint source	Why it is that number
Length × Width	1200 × 700 mm	published by this proposal	Back-calculated from the standalone kerb cut width in FIG.12: a device that cannot pass the opening is not accepted by this proposal
Height (with load)	1300 mm	Set by this proposal	Taken below the sightline of a standing adult, so that queuing devices do not block eye contact across the sidewalk; no external basis, set by this proposal
Turning diameter	1800 mm	Set by this proposal	Required to turn in place; it sets the shape of the reservoir space and not just its area — precisely the half that "18 m ² " cannot state
Kerb weight	≤ 120 kg	Set by this proposal	Taken at the order a single person can push clear after a power failure; beyond that, mechanical assistance is required, and there is nowhere to place it at a kerb cut
Site speed	≤ 6 km/h	Set by this proposal	Speed ceiling in the shared pedestrian-vehicle zone, taken at the order of walking speed; no citable local regulation was obtained, so it is registered as a self-set value
Clear width left after passing	≥ 1500 mm	published by this proposal	Step-free clear walking width, the same value as KIT-04; equipment passage may not press it below this number
Marking	Cap height ≥ 25 mm	published by this proposal	ID number, responsible body and appeal channel, all three present; same rule as the reading plate: attribution starts with a visible ID number (FIG.16, FIG.20)

4. Converting 18 m² into a number of units — the two drawings must agree with each other

One standing bay is the envelope length times the turning circle: 2.16 m². FIG.12 publishes a 18 m² floor for the reservoir, so that is 8 at once. The number did not exist before: "≥ 18 m²" never said how many, and two devices of equal footprint with different turning radii do not need the same reservoir. The build checks the two sheets against each other — if the area does not divide into bays, or the count exceeds what the floor holds, it refuses.

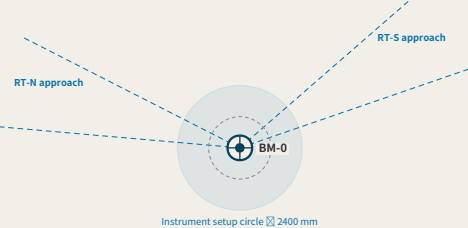
FIG.12 fixed the device reservoir at ≥ 18 m² and this package had never drawn a device. Area is not specification: equal footprints with different turning radii need different reservoirs. An envelope, not a product; 7 dimensions each marked with what bounds them, 4 bounded by nothing but this proposal — attack those first. It holds 18 m² / 8 devices.

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BM-0, the origin: the place both routes have to return to

FIG.25

1. Datum site: instrument setup circle, clear walking width, approach sightlines of the two routes



2. Closure record: readable without

A reading is published the moment it is taken, yet the evidence is public only as far as the door. So the closure record sits outside the door, facing the sidewalk, on the plate geometry FIG.16 fixes (eye height 900–1350 mm). Inside the hall it would turn anyone can walk up and read into during opening hours.

3. The six requirements the datum must meet, and the source of each

Item	Requirement	Why, and where it comes from
Instrument setup circle and the step-free clear width	Diameter 2400 mm ≥ 1500 mm clear outside the setup circle	Tripod spread 1,200 mm, operator outer circle 600 mm; closing a route means setting the instrument up on this stone, and this circle is the ground it actually takes Instrument setup may not eat the clear width — FIG.24' s rule for equipment, no slack for our own people
Route intervisibility	One approach sight cone each; no trees or parking inside	RT-N and RT-S terminate here, which needs sight of the previous turning point; without it you add one, and each adds a height difference
Closure record orientation	Faces the sidewalk, eye height 900–1350 mm, read without entering	Published as taken, but evidence is open only as far as the door — a datum with opening hours is shut when the reading lands
Display content	Network closure error $\geq 1\%$ out	Network closure error only; a single-point failure shows there (erratum E50). Moving bad news back to the datum is the recorded defect
Fall	$\geq 1\%$ out	As FIG.22: the stone must be a local high point, or in January it is the only ice here

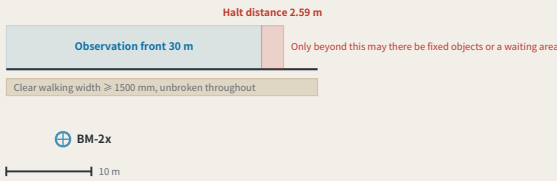
BM-0 appears on nearly every sheet here and had never been drawn — and what it carries is a spatial property: a reading is published the moment it is taken, and evidence is only as public as the door. Hang the record in a hall with opening hours and the origin is shut at the moment the reading lands. This sheet gives the setup circle, an approach sightline cone for each closing route, and the closure record outside the door facing the sidewalk.

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BM-2x, the second order: a scenario has to be stoppable, and that takes ground

FIG.26

1. Observation front and halt distance, same scale



2. This distance is calculated, not decreed

FIG.24 fixes the site speed at ≤ 6 km/h = 1.667 m/s. Halt distance = reaction leg + deceleration leg = 1.667 × 1 + 1.667² ÷ (2 × 1.5) = 1.67 + 0.93 = **2.59 m**. The reaction time and the deceleration are both set by this proposal, not code values — logged in the register format as "A-DEVICE-002", with the derivation, the consequences of a wrong value, the closing condition and the ownership, submitted together with this drawing. The formula is published with the drawing: change one set of values and this stretch of ground can be recomputed at once.

3. Where second-order and third-order points differ

Item	Value	Why
stone	Same KIT-01 as the third-order point	Second order and third order differ in re-survey interval and route role, not in construction — a more important place gets no special benchmark
Re-survey cycle	Quarterly	The two wings are the source of scenario density; readings are taken quarterly, priced on the same table as the annual first order and the monthly third order
Observation front	30 m	The stretch of kerb the scenario is actually watched from. "The scenario takes its reading here", given without a length, is the same kind of sentence as "reservoir space 18 m" given without devices
Halt distance	2.59 m	From FIG.24: a site speed of ≤ 6 km/h gives reaction 1 s over 1.67 m, then deceleration at 1.5 m/s² over 0.93 m
Clearance requirement	No fixed objects and no waiting area within ≥ 2.59 m ahead of the front	A scenario that cannot stop is not a scenario, it is an accident; this stretch of ground is the physical form of the words "it can stop"
Clear walking width	≥ 1500 mm, uninterrupted throughout	The same number as in FIG.24 and FIG.25 — observation, instrument setup, equipment passage: none of them may eat into it

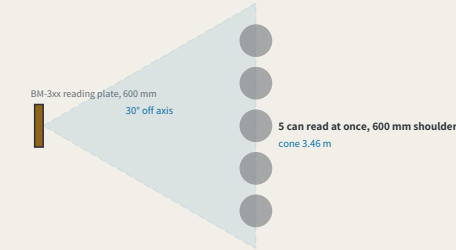
The two wings carry the second-order points; they appear in the naming table, the cost table and the resource table, and have never been drawn. What separates second order from third order is not the stone — both are KIT-01 — but that what runs here is a **scenario aimed at real users**, so it must be able to stop. Computed from FIG.24 at ≤ 6 km/h, the halt distance is 2.59 m; within that stretch of ground ahead of the front there may be no fixed objects and no waiting area. The reaction time and the deceleration are both set by this proposal and are registered as A-DEVICE-002.

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Third order: the most numerous tier, and more than one reader

FIG.28

I. The legible cone, and how many can read at once



II. The part ground cannot solve

A class of 30 cannot read at once. 5 fit the cone, so 6 rounds, and at 40 s a reading that is about 4 minutes. The number is here because widening the ground does not solve it. No cone holds thirty people; the rest take turns, and turns cost time — a cost that, unwritten, gets discovered on site in the form of "this point is awkward". The off-axis angle and the shoulder width are chosen by this proposal and registered as "A-READ-001"; the plate width, the viewing distance and the clear width come from FIG.16, FIG.27 and FIG.24 and are not fixed here.

clear width 1500 mm — a queue runs parallel to the kerb, never pushing into this

III. Standing rules for a third-order point

Item	Value	Why
Re-survey cycle	Monthly	third order is the most numerous tier and the one residents actually stand at
Legible cone	3.46 m	a 600 mm plate (FIG.16) stays legible within 30° off axis, and at the distance FIG.27 fixes, 3.0 m, the cone opens to this width
Readers at once	5	cone ÷ a 600 mm shoulder. This is this sheet answering the question FIG.27 left open — that sheet fixed one person's position
A class of thirty	6 rounds, about 4 minutes	past that count the extra people are not handled by ground but by taking turns — and turns are a time cost, which belongs in writing rather than being discovered on site
Which way a queue runs	parallel to the kerb, never perpendicular to the direction of travel	a perpendicular queue pushes its tail into the 1500 mm clear width; a parallel one does not
What is not provided	no railing, no dedicated waiting area	a third-order point sits in a neighbourhood, and fencing it stops it being something you can read in passing; what this sheet gives is a standing rule, not a new structure

The origin and the second order each have a sheet; the third order is the tier nobody drew, and the one residents actually stand at. It also inherits what FIG.27 left open: that sheet's 3.0 m distance is one person's position. From a 600 mm plate and a 30° off-axis limit the cone is 3.46 m wide and 5 people read at once; a class of 30 takes 6 rounds, about 4 minutes — that part is solved by turns, not by ground.

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One reading: where a person stands, and whether they block the way

FIG.27

a type drawing; it carries no orientation

I. Where the person reading stands



clear width 1500 mm — the person reading may not stand inside it

II. This viewing distance comes from a ratio this package already published

The body text long ago checked its own drawings against "legible cap height ≈ viewing distance ÷ 250", and found the minimum cap height across all drawings, 4.73 mm, compliant. That same ratio has never been pointed at the plate face. Pointed at it, what it returns is not a cap height but a standing position: cap height 12 mm, viewing distance 3.0 m, and 3.0 m lies outside the 1500 mm clear walking width — the person reading and the person walking past neither yield to nor obstruct each other. Take a cap height of 6 mm and the viewing distance becomes 1.5 m, putting the reader squarely in the middle of the path: the plate face is still legible, the sidewalk no longer works. The 12 mm cap height is set by this proposal and is registered as "A-PLATE-001". This package carries a second cap height: FIG.24 requires device markings ≥ 25 mm, legible at 6.25 m by the same ratio — not a contradiction, and not an accident: device markings exist to identify the machine coming toward you, and must be read at a distance that still leaves time to react; the reading plate is read standing in front of it. Different distances, different cap heights — and with no stated relation between the two, a reader will simply assume one of them was set arbitrarily. The ratio 250 is not self-set; it is the one this package already applies to itself.

III. The four steps of one reading, and the ground each takes

Step	What happens	Why that dimension
Walk there	FIG.21 measured the worst walk at 1,111 m — 30.9 minutes for PS — and this proposal does not meet its own rule	The cost of this step has been measured, and it measures as non-compliant
Stand	Reading at 3.0 m from the plate face, standing outside the clear walking width	Cap height 12 mm × viewing distance ratio 250 = 3.0 m; take the cap height as 6 mm and the viewing distance drops to 1.5 m, putting the person reading in the middle of the path
Read	Read the number on the plate and the benchmark_id	The number makes this reading attributable; no need to ask afterwards where the person was standing
Disagree	Scanning the appeal code requires approaching to about 600 mm	This step has to come within close range, so the stacking goes on the side facing away from the direction of travel — the one step that must intrude intrudes on the edge of the path, not its centre

The whole participation argument rests on a single sentence — anyone can walk up to a benchmark and take a reading — yet no drawing has ever drawn the four minutes that person spends standing there. Standing is not free: the ground he occupies is the ground others walk on. Working backwards from the "legible cap height ≈ viewing distance ÷ 250" already published in this package, a cap height of 12 mm gives a viewing distance of 3.0 m, which falls just outside the 1500 mm clear walking width; take 6 mm and the reader stands in the middle of the path. The cap height is set by this proposal and registered as A-PLATE-001.

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The annual first-order closure: how long a route actually takes

FIG.29

I. The two closing routes, and how many setups each takes



Each vertical line is a setup, spaced by the 50 m sight — the instrument stands midway between two staves, so one setup advances the run by twice the sight. A closure is not one act but this many, and each is a place it can go wrong, which is exactly what the closure error measures.

1,000 m

II. The duration derived from geometry, against the band the cost table publishes

Route	Length	Instrument setup	Observing	Walk back	Total
RT-N	5402 m	55	4.6 h	1.3 h	5.8 h
RT-S	5177 m	52	4.3 h	1.2 h	5.5 h

the band the cost table publishes for a first-order session: 4.0-6.0 h

III. These two numbers had never met

The 4.0-6.0 h for a first-order point in "operations.json" was written into the model by hand, on its author's judgement, and nothing could check it. The route lengths are not hand-written: they come out of the shipped geometry. Divide the length by the viewing distance for the instrument setups, multiply by the minutes each takes, add the walk back to the datum along the spine, and the result is 5.5-5.8 h — inside that band, and tight against its upper edge. The build asserts it: lengthen the geometry or edit the band and the two stop being compatible and the build stops. The viewing distance and the minutes per setup are chosen by this proposal and registered as "A-SURVEY-001"; the route lengths, the walking speed and the hour band are all already published here.

The first order was the last tier without a sheet, and it is not really a place but an event: once a year a closing route is run, and this proposal's measure of trust comes out of that run. From the shipped route lengths, a 50 m sight and 5 minutes a setup, the two routes need 52-55 setups and 5.5-5.8 h — inside the 4.0-6.0 h the cost table sets by hand — and the build asserts the two stay compatible.

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The year: where forty-seven readings fall

FIG.30

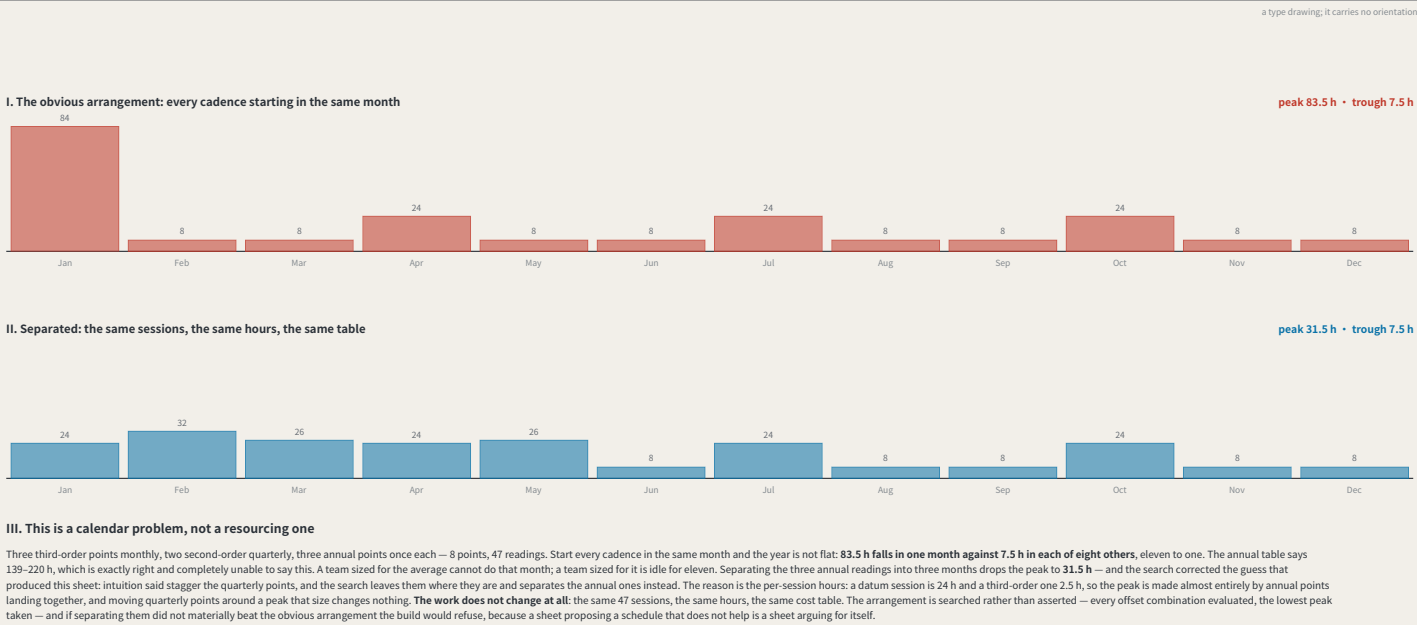


FIG.29 priced **one** closure and the cost table prices a **year** — and no sheet had drawn the year. Drawn, it shows that with every cadence starting in the same month the peak is 83.5 h against a 7.5 h trough, eleven to one: a team sized for the average cannot do the peak month, and one sized for the peak is idle the rest of the year. Separating the three annual readings drops it to 31.5 h with **no change to the work at all**. This is a calendar problem, not a resourcing one.

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The other year: the hours this network asks of people it does not pay

FIG.31



FIG.30 flattened **paid** hours and said the quarterly points need not move. Volunteers attend only the second and third order — precisely the tier left alone — so the volunteer year stayed at 31.0 h against 15.0 h. Separating them brings it to 23.0 h at a cost of 2.0 h on the paid peak. Both cannot be minimised at once, and this proposal takes the volunteer peak: a paid peak is a procurement problem, a volunteer peak is a participation failure.

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How many people: the cheapest roster is the one that empties the instrument

FIG.32

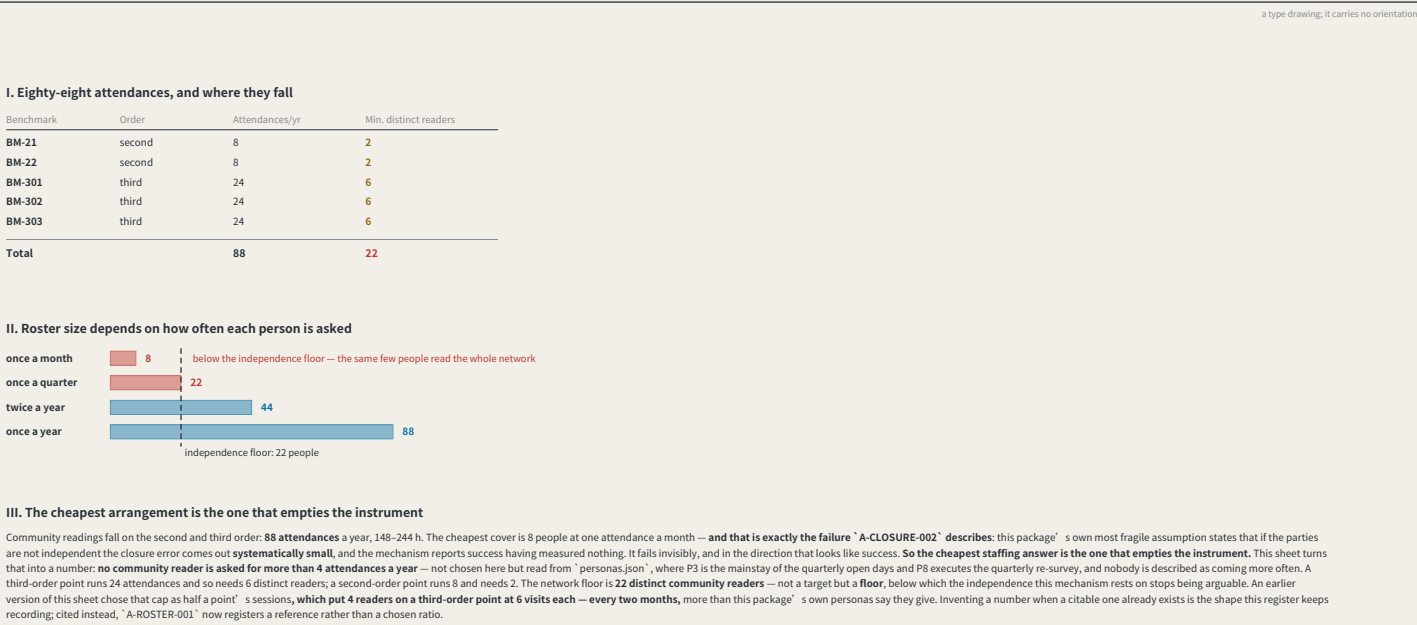
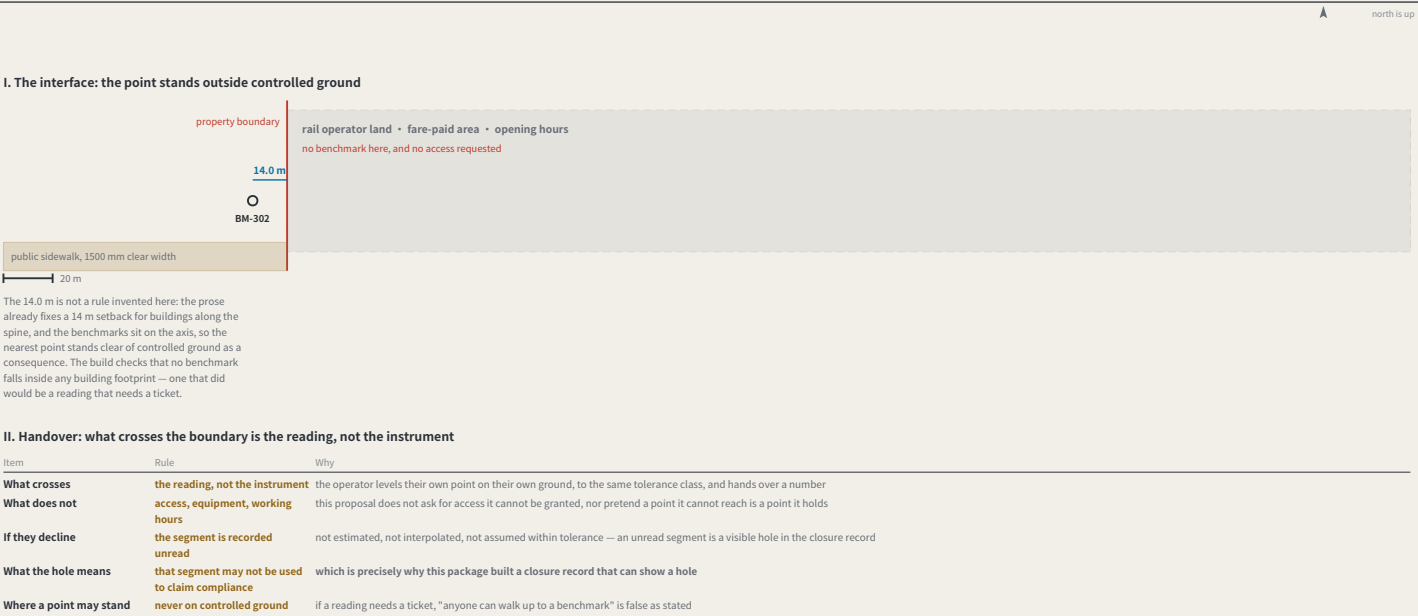


FIG.31 scheduled volunteer **hours** and never asked how many **people** that is. The year needs 88 community attendances, and the cheapest cover is 8 people once a month — **which is exactly the failure A-CLOSURE-002 sets out**: parties that are not independent make the closure error systematically small, and the mechanism reports success having measured nothing. The rule: nobody attends more than half a benchmark’ s readings in a year, giving a network floor of 22 people.

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Dazhongsi: where the belt meets ground this network does not own

FIG.33



Every other sheet assumes this network reaches its own points; at the station that ends — the landowner is not the city. Three rules: **no benchmark on controlled ground** (BM-302 stands 14.0 m clear of the hub, checked at build time); **what crosses is the reading, not the instrument**; and **if the operator declines the segment is recorded unread** — not estimated, not interpolated, a visible hole in the closure record.

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